

A KEY ROLE FOR OREXIN IN PANIC ANXIETY

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Panic disorder is a severe anxiety disorder with recurrent, debilitating panic attacks. In individuals with panic disorder there is evidence of decreased central gamma-aminobutyric acid (GABA) activity as well as marked increases in autonomic and respiratory responses after intravenous infusions of hypertonic sodium lactate. In a rat model of panic disorder, chronic inhibition of GABA synthesis in the dorsomedial-perifornical hypothalamus of rats produces anxiety-like states and a similar vulnerability to sodium lactate-induced cardioexcitatory responses. The dorsomedial-perifornical hypothalamus is enriched in neurons containing orexin (ORX, also known as hypocretin), which have a crucial role in arousal, vigilance and central autonomic mobilization, all of which are key components of panic. Here we show that activation of ORX-synthesizing neurons is necessary for developing a panic-prone state in the rat panic model, and either silencing of the hypothalamic gene encoding ORX (*Hcrt*) with RNAi or systemic ORX-1 receptor antagonists blocks the panic responses. Moreover, we show that human subjects with panic anxiety have elevated levels of ORX in the cerebrospinal fluid compared to subjects without panic anxiety. Taken together, our results suggest that the ORX system may be involved in the pathophysiology of panic anxiety and that ORX antagonists constitute a potential new treatment strategy for panic disorder.